Z+T Ch 3 9-9

Section 4: Read

Section 5 Conservation Laws

$$a(z)\, z_x + z_y = 0$$

assume $a(z), f(x) \in C^1$

I.C. $z(x,0) = f(x)$

y represents time

ex) Conservation of mass (not responsible)

ρ = density $\quad \underline{v}$ = velocity

$\rho \underline{v}$ = mass flux $= q(\rho)$

$$\frac{\text{mass}}{L^3}\,\frac{L}{\text{time}} = \frac{\text{mass}}{L^2\text{-time}}$$

mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}\left(q(\rho)\right) = 0$$

$$\frac{\partial \rho}{\partial t} + \frac{dq}{d\rho}\frac{\partial \rho}{\partial x} = 0$$

$t \to y$

$\rho \to z$

$$\frac{\partial z}{\partial t} + a(z)\frac{\partial z}{\partial x} = 0$$

① ~~P4~~ $Pz_x + Qz_y = R$

$P = a(z) \quad Q = 1 \quad R = 0$

$x = \tau \quad y = 0 \quad z = f(\tau)$

② check for existence + uniqueness

(i) P, Q, R, $x_0(\tau)$ $y_0(\tau)$ $z_0(\tau) \in C^1$ Yes ✓

(ii) Is the I.C. characteristic?

$$\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0' & y_0' \end{vmatrix} = \begin{vmatrix} a(f(\tau)) & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0 \text{ Good!}$$

③ charastic curves:

$\frac{dx}{ds} = P \qquad \frac{dy}{ds} = Q \qquad \frac{dz}{ds} = R$

$x(s=0) = x_0(\tau) \qquad y(s=0) = y_0(\tau) \qquad z(s=0) = z_0(\tau)$

$$\frac{dx}{ds} = a(z) \qquad \frac{dy}{ds} = 1 \qquad \frac{dz}{ds} = 0$$

$$x_0(s=0) = \tau \qquad y_0(s=0) = 0 \qquad z(s=0) = f(\tau)$$

Treat τ as a constant

$$\frac{dx}{ds} = a(f(\tau))$$

$$x = a(f(\tau))s + c_1$$

$$s=0,\ x=\tau$$

$$\tau = 0 + c_1 \Rightarrow c_1 = \tau$$

$$\underline{x = a(f(\tau))s + \tau}$$

(1)

$$dy = ds$$

$$y = s + c_2$$

$$s=0,\ y=0$$

$$0 = 0 + c_2 \Rightarrow c_2 = 0$$

$$\underline{y = s}$$

(2)

$$z = c_3$$

when $s=0$ $z = f(\tau)$

$$f(\tau) = c_3$$

(ok! τ is a constant.)

$$\underline{z = f(\tau)}$$

(3)

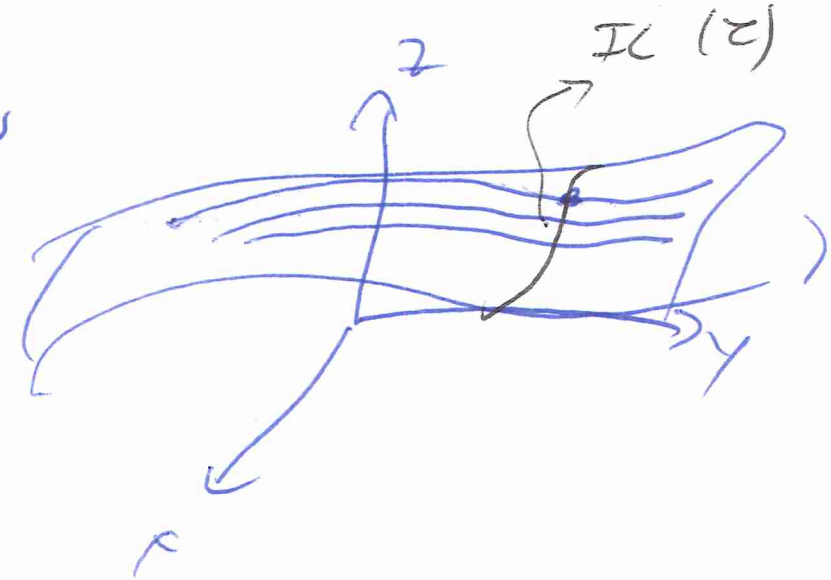
- Look at characteristics.

eliminate s

$\cancel{x}$ ① + ②

② $s = y$ $\xrightarrow{\text{subst}}$ ① $x = a(f(\tau))\,y + \tau$

Along ~~each~~ each charact. curve, τ is fixed.



$z = f(\tau) = z_0$

I.C. $\tau = \text{constant} = x_0$

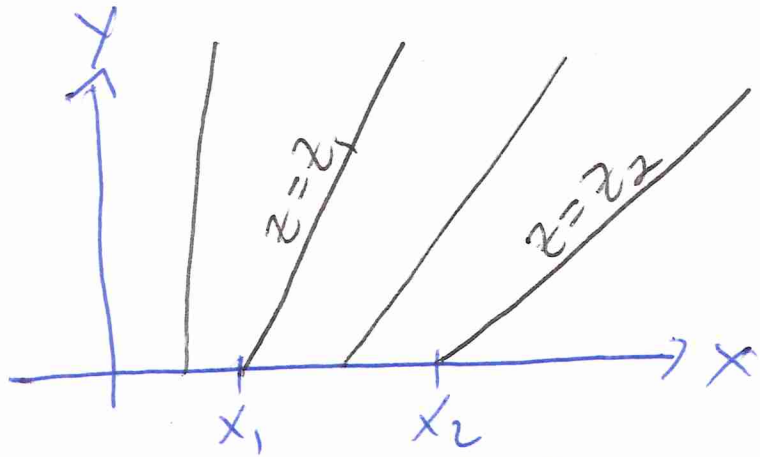
$x = a(z_0)\,y + x_0$

$x - x_0 = a(z_0)\,y$

$$y = \frac{1}{a(z_0)}(x - x_0)$$

lines with slopes $\frac{1}{a(z_0)}$

(i) $a(z)$ is ~~decreasing~~ increasing

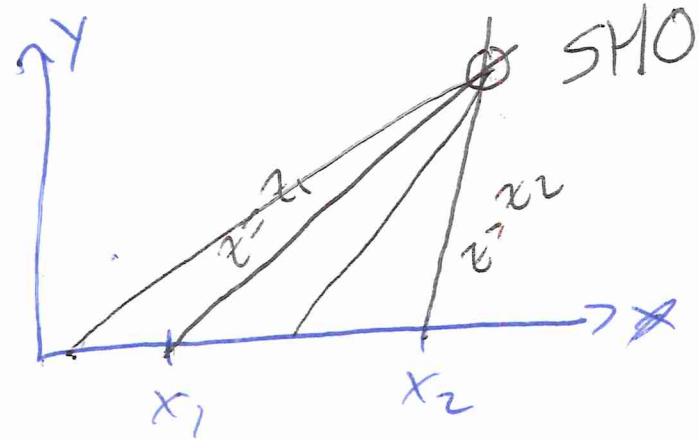


$a(z_1) < a(z_2)$

$\frac{1}{a(z_1)} > \frac{1}{a(z_2)}$

$a(z)$

(ii) $a(z)$ is decreasing



$a(z_1) > a(z_2)$

$\frac{1}{a(z_1)} < \frac{1}{a(z_2)}$

Get a shock unless

$z_1 = z_2$

or $f(x_1) = f(x_2)$

(ii) Look at sol'n

eliminate both s + τ $\quad z = f(x,y)$

$x = a(f(\tau))y + \tau$ $\quad z = f(\tau)$ (eliminated s)

$\tau = f^{-1}(z)$

$$x = a(z)y + f^{-1}(z)$$

$$f^{-1}(z) = x - a(z)y$$

$$z = f(x - a(z)y)$$ implicit sol'n!

How do we know where $z = f(x,y)$ is a f'n? $\Rightarrow$ OK

<u>implicit f'n Thm!</u> $F(x,y,z) = z - \overset{f(x,y,z)}{f(x,y)} = 0$ Level surface

IFT $\Rightarrow$ we have a f'n as long as

$$\frac{\partial F}{\partial z} \neq 0$$

$$F(x,y,z) = z - f(x - a(z)y) = 0$$

$$\left.\frac{\partial F}{\partial z}\right|_{x,y} = 1 - f'(x - a(z)y)(-a'(z)y) \neq 0$$

AT I.C. $y = 0$ $\quad F_z = 1 > 0$

want $F_z > 0$

our implicit sol'n is good as long as

$$F_z = \underline{1 + f'(x - a(z)y)\, a'(z)\, y} > 0$$

same

IFT $$z_x = \frac{f'(x - a(z)y)}{\underline{1 + f'(x - a(z)y)\, a'(z)\, y}}$$

So as $F_z \to 0$ $\quad z_x$ (and z_y) $\to \pm\infty$

Summary for conservation law of the form

$$a(z)z_x + z_y = 0 \qquad z(x,0) = f(x)$$

sol'n is given implicitly by

$$z = f(x - a(z)y)$$

characteristics : $x - x_0 = a(z_0)y$

& on charact. $z = f(x_0)$

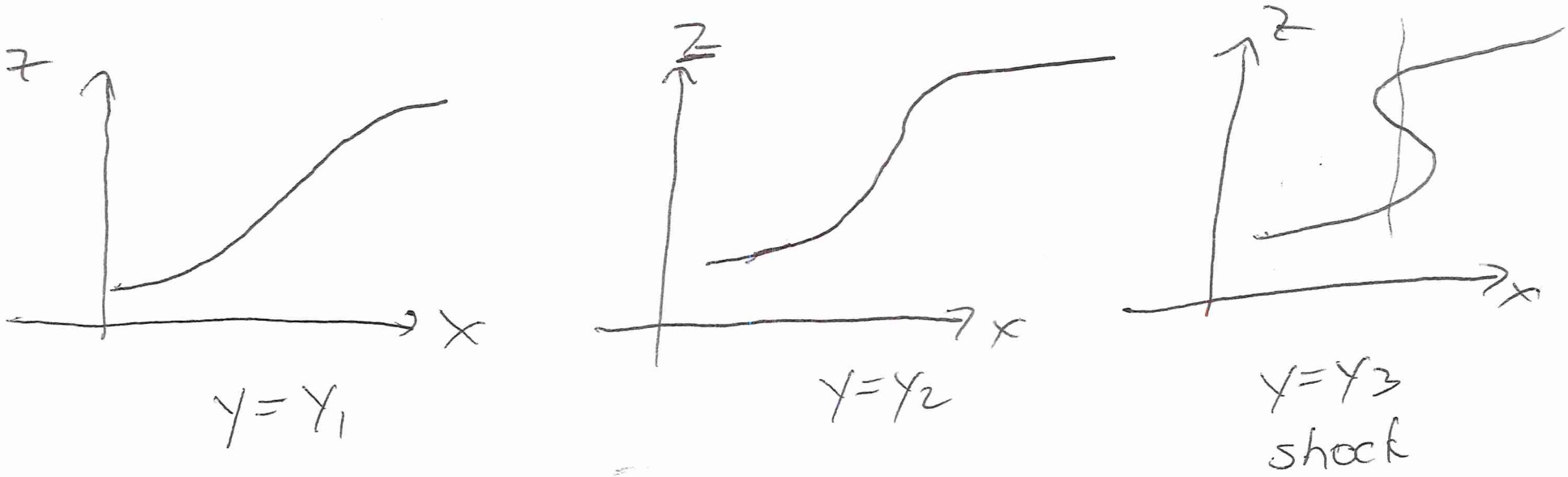
condition for $z = f(x,y)$ (function)

$$1 + f'(x - a(z)y)a'(z)y > 0$$

If $= 0$, then $z_x, z_y \to \pm\infty$ & $z \notin C^1$

& sol'n develops a shock.

Usually what happens at shock (not responsible)



$y = y_1$ $\quad$ $y = y_2$ $\quad$ $y = y_3$

shock

mathematically:

add an additional condition (entropy condition) which tells us which soln is the path the system will take.